Nicolas Stevens

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Professional Experience

2026 – present **Postdoctoral Researcher**, *Inria Center of the University of Lille*, Lille, France

2025 – 2026 **Postdoctoral Researcher**, *Max Planck Institute*, Bonn, Germany
Research under the supervision of Prof. Peter Cramton. Funded by Forward Market Design.

Fall 2024 **Visiting Researcher**, *Harvard Kennedy School*, Cambridge, United States
Research under the supervision of Prof. William Hogan. Invited speaker at MIT's LIDS EIMC2 Seminar and Harvard's Energy Policy Seminar.

2020 – 2024 **PhD Candidate**, *CORE, UCLouvain*, Belgium
Thesis: "*Price Formation with Non-Convexities: Theory and Applications for the Electricity Market*". Supervised by Prof. Anthony Papavasiliou. Relevant coursework: robust optimization, advanced microeconomics I and II, social choice theory.

2019 – 2020 **Senior Consultant in Optimization and Energy Markets**, *N-SIDE*, Belgium

2016 – 2019 **Consultant in Optimization and Energy Markets**, *N-SIDE*, Belgium
Applied mathematical optimization to problems in the energy sector. Working with various stakeholders (transmission and distribution system operators, power exchanges). Responsibilities included developing optimization prototypes, designing mathematical models and market mechanisms, managing project teams, client relationships, and business development activities. Selected projects:
- *Elia: Dynamic dimensioning of operating reserves (2016-2017).*
- *European power exchanges: EUPHEMIA market-clearing algorithm support (2017-2018).*
- *Elia: Optimization tool for maintenance planning (2018).*
- *Statnett: balancing market study (2019-2020).*
- *Indian Energy Exchange (IEX): designing the day-ahead market clearing algorithm (2019).*
- *European Commission – CoordiNet (Horizon 2020): task leader for a project on local energy markets (2019–2020).*

Education

2024 **PhD in Applied Mathematics**, *UCLouvain*

2023 **MA in Philosophy**, *UCLouvain*, *Summa cum laude*

2020 **University Certificate in Philosophy**, *UCLouvain*, *Magna cum laude*

2016 **MSc in Engineering (Applied Mathematics)**, *UCLouvain*, *Magna cum laude*
Master thesis: *Models and Algorithms for Pricing Electricity in Unit Commitment*. Relevant coursework: operations research, stochastic optimisation, optimisation in production scheduling, financial mathematics, non-linear programming.

Fall 2015 **Master's Exchange Programme**, *Polytechnique Montréal*

2014 **BSc in Engineering (Mechanics & Applied Mathematics)**, *UCLouvain*, *Magna cum laude*
Relevant coursework: graph theory, game theory, optimisation.

Publications

Journal Articles Stevens, N., O'Neill, R., and Papavasiliou, A. (2026). Average incremental cost pricing in electricity auctions. *Energy Economics*, 153, 109047.

Ponthiere, G., and Stevens, N. (2025). The Morality of Markets. A Critique. *Mathematical Social Sciences*, 134: 14-19

Stevens, N., Papavasiliou, A., and Smeers, Y. (2024). On some advantages of convex hull pricing for the European electricity auction. *Energy Economics*, 134, 107542.

Stevens, N., Smeers, Y., and Papavasiliou, A. (2024). Indivisibilities in investment and the role of a capacity market. *Journal of Regulatory Economics*, 66, pp. 238–272.

Stevens, N., and Papavasiliou, A. (2022). Application of the Level Method for Computing Locational Convex Hull Prices. *IEEE Transactions on Power Systems*, 37(5), 3958-3968.

Mezghani, I., Stevens, N., Papavasiliou, A., and Chatzigiannis, D. I. (2023). Hierarchical coordination of transmission and distribution system operations in European balancing markets. *IEEE Transactions on Power Systems* 38 (5), 3990-4002.

De Vos, K., Stevens, N., Devolder, O., Papavasiliou, A., Hebb, B., and Matthys-Donnadieu, J. (2019). Dynamic dimensioning approach for operating reserves: Proof of concept in Belgium. *Energy Policy*, 124, 272-285.

Conference Papers

Papavasiliou, A., Bjørndal, M., Doorman, G., and Stevens, N. (2020). Hierarchical Balancing in Zonal Markets. *17th International Conference on the European Energy Market (EEM)*, pp. 1-6, IEEE.

Absil, P. A., Sluysmans, B., and Stevens, N. (2018). MIQP-based algorithm for the global solution of economic dispatch problems with valve-point effects. *Power Systems Computation Conference (PSCC)*, pp. 1-7, IEEE.

Working Papers

Stevens, N., Cramton, P. and Toniotti, M. (2026) Does Financial Trading Smooth Non-Convex Markets? arXiv:2607.06316

Awards and Grants

2026 FNRS Postdoctoral (CR) Fellowship. Three-year postdoctoral fellowship. Scheduled to commence in October 2027. Title: "Optimization and market design for flexibility in electricity markets".

2026 PGMO IROE Fellowship. One-year postdoctoral fellowship. Title: "From Complementarity to Multi-Agent Learning in Electricity Markets"

2024 Travel grants: WBI-World fellowship

2016 IEEE ICTEAM Best Master's Thesis Award, Second Prize

Teaching Experience

Lecturer

Quantitative Energy Economics, 2022–2023, 2023–2024 (5 ECTS, UCLouvain/EPL: LINMA2415, graduate course, ∼25 students, Prof. Gauthier de Maere). Delivered part of the course lectures.

Energy Markets and Demand, 2025 (5 ECTS, UCLouvain/LSM: LLSMS2052, graduate course, ∼20 students, Prof. Willems). Guest lecturer (2-hour lecture).

CEER Training Academy Programme, 2025 & 2026. One-hour training session for energy-regulation professionals and academics.

Teaching Assistant

Combinatorial optimization, 2021–2022, 2022–2023 (5 ECTS, UCLouvain/EPL: LINMA2450, graduate course, ∼35 students, 2h/week, Prof. Hendrickx and Prof. Grapiglia)

Project in applied mathematics, 2020–2021 (5 ECTS, UCLouvain/EPL: LFSAB1507, undergraduate course, ∼30 students, 2h/week, Prof. Papavasiliou)

Quantitative Energy Economics, 2022–2023, 2023–2024 (5 ECTS, UCLouvain/EPL: LINMA2415, graduate course, ∼25 students, 2h/week, Prof. Gauthier de Maere)

Economics (for engineers), 2021–2022 (5 ECTS, UCLouvain/EPL: LEPL1803, undergraduate course, ∼30 students, 2h/week, Prof. Hendrickx)

Mentoring

Master's thesis committee member (2024): Anthony Chung.

Selected Talks

2026 EWI energy economics seminar (Cologne), INRIA Inocs seminar (Lille), CORE 60 young researcher workshop (LLN)

2025 IAEE conference (Paris), N-SIDE internal seminar (industry talk), KULeuven ESIM seminar (Leuven)

2024 Engie internal seminar (industry talk), IAEE conference (Istanbul), Iceberg ERC workshop (Athens), Harvard Energy Policy Seminar (Cambridge), Harvard Climate Research Scholars Breakfast (Cambridge), MIT LIDS EIMC2 Seminar (Cambridge)

2023 CORE doctoral workshop (LLN), FSR Loyola Autumn Research School (Florence), INFORMS annual meeting (Phoenix)

2022 EURO conference (Espoo), INFORMS annual meeting (Indianapolis)

2021 EURO conference (Athens), INFORMS annual meeting (Anaheim)

Miscellaneous

Reviewing activities EURO Journal on Computational Optimization, Operations Research Perspectives, IEEE Transactions on Power Systems, The Energy Journal, Energy Economics, IEEE Transactions on Energy Markets, Policy, and Regulation.

Languages French (native speaker), English (professional and fluent), Spanish (elementary level), Dutch (elementary level)